FUNGIBLE

OUR GOAL:

Revolutionize the Performance, Reliability, and Economics of all Data Centers

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Fundamental Beliefs:

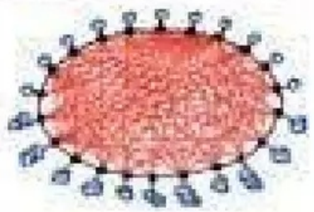
- Human progress ∝ Information technology

- Insatiable demand for:
  - Compute
  - Storage
  - Networking



- AI will drive the compute model

- Scale-out will be the dominant architecture



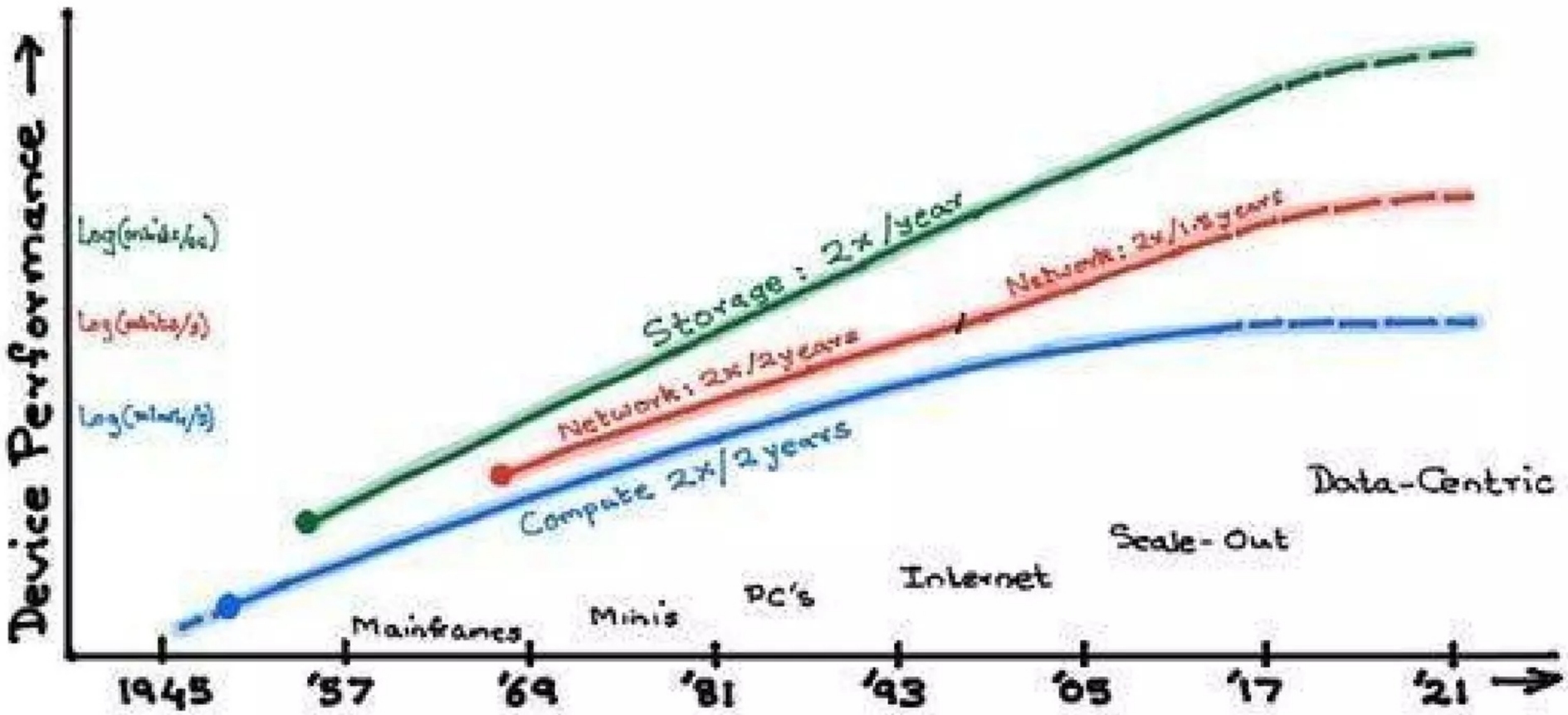
Network is key

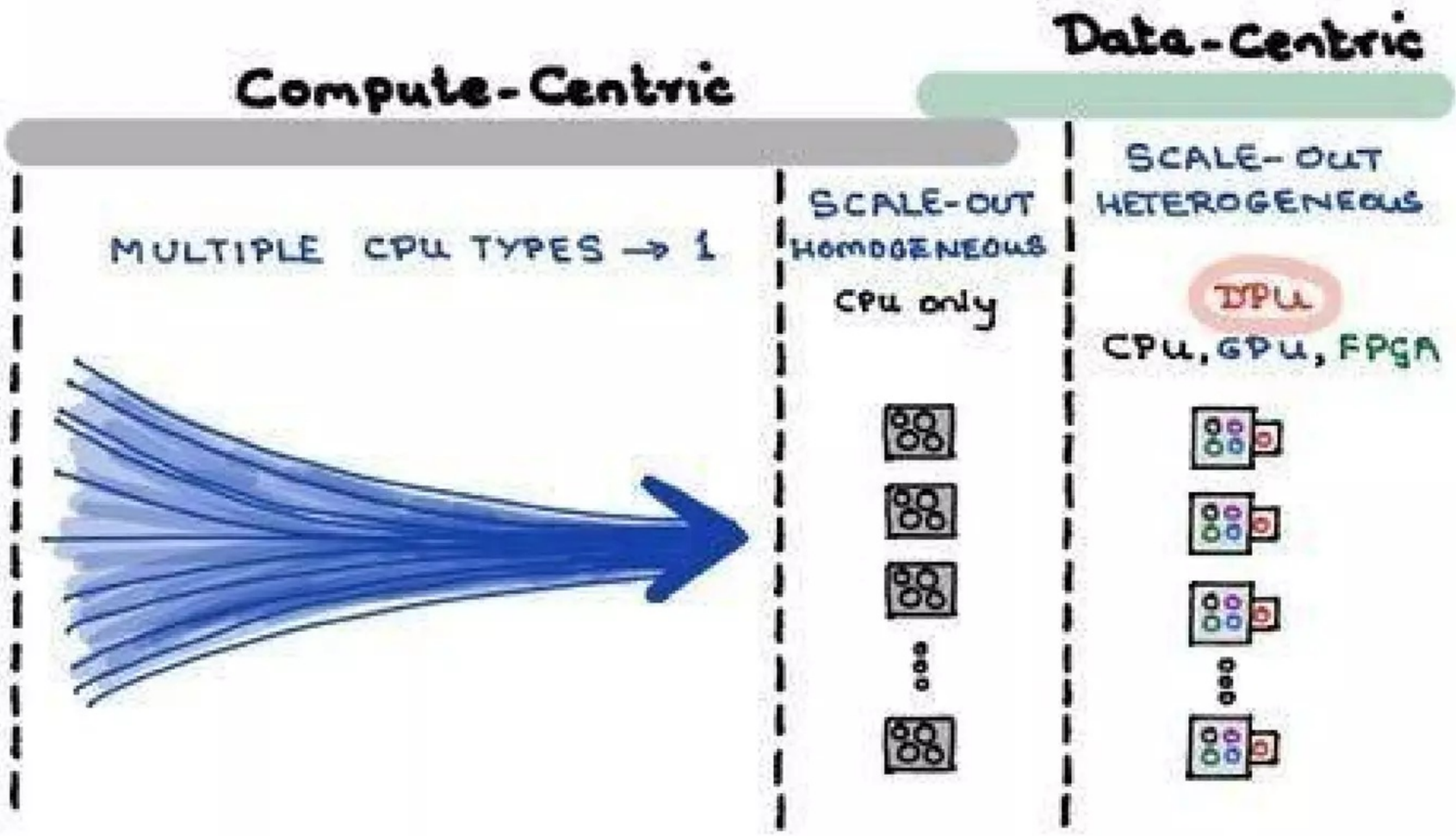


- Cloud will be the dominant delivery model
  - Agility
  - Economics

KEY INDUSTRY TRENDS:

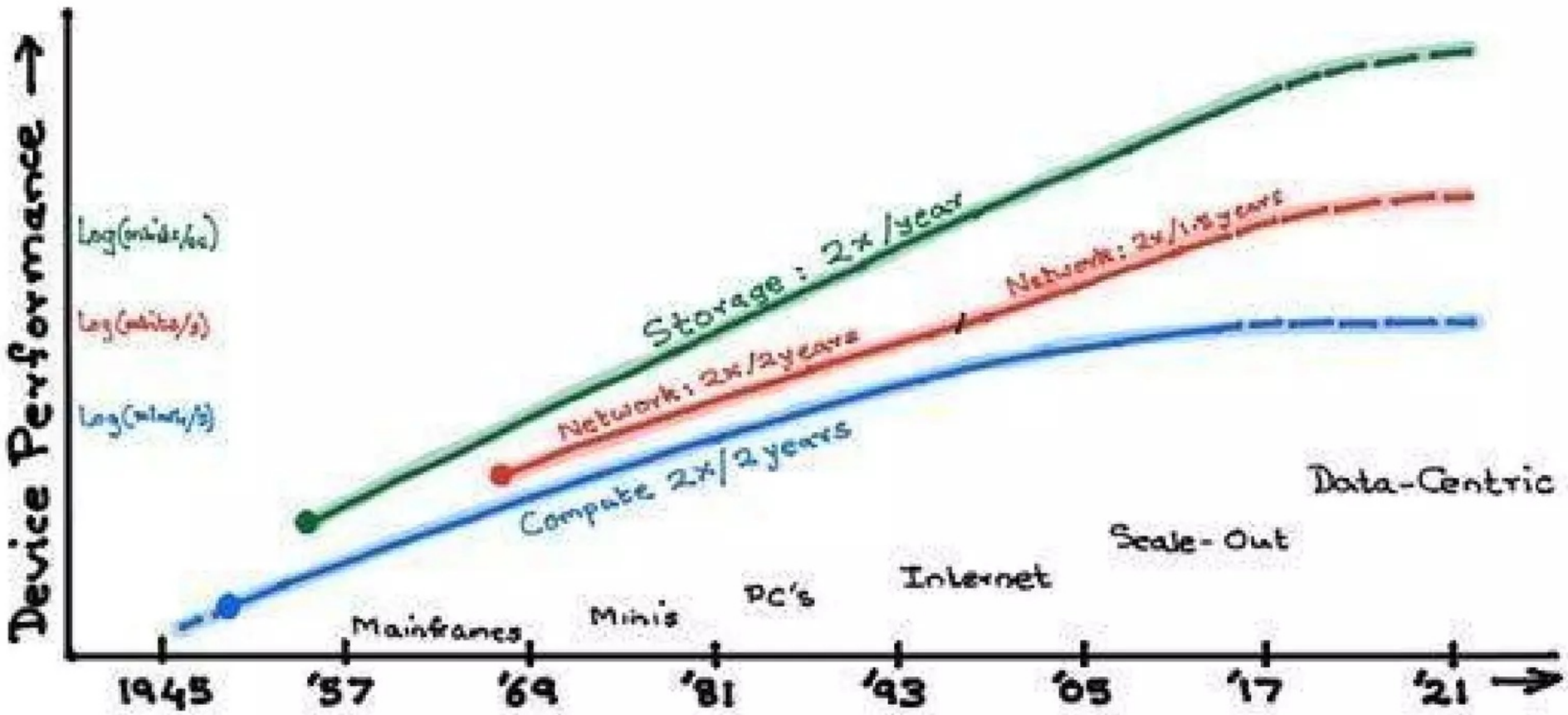
- Moore's law plateaus
- Hyper connectivity is pervasive
- Apps become increasingly Data Centric (AI, Analytics)
- Security threats accelerate

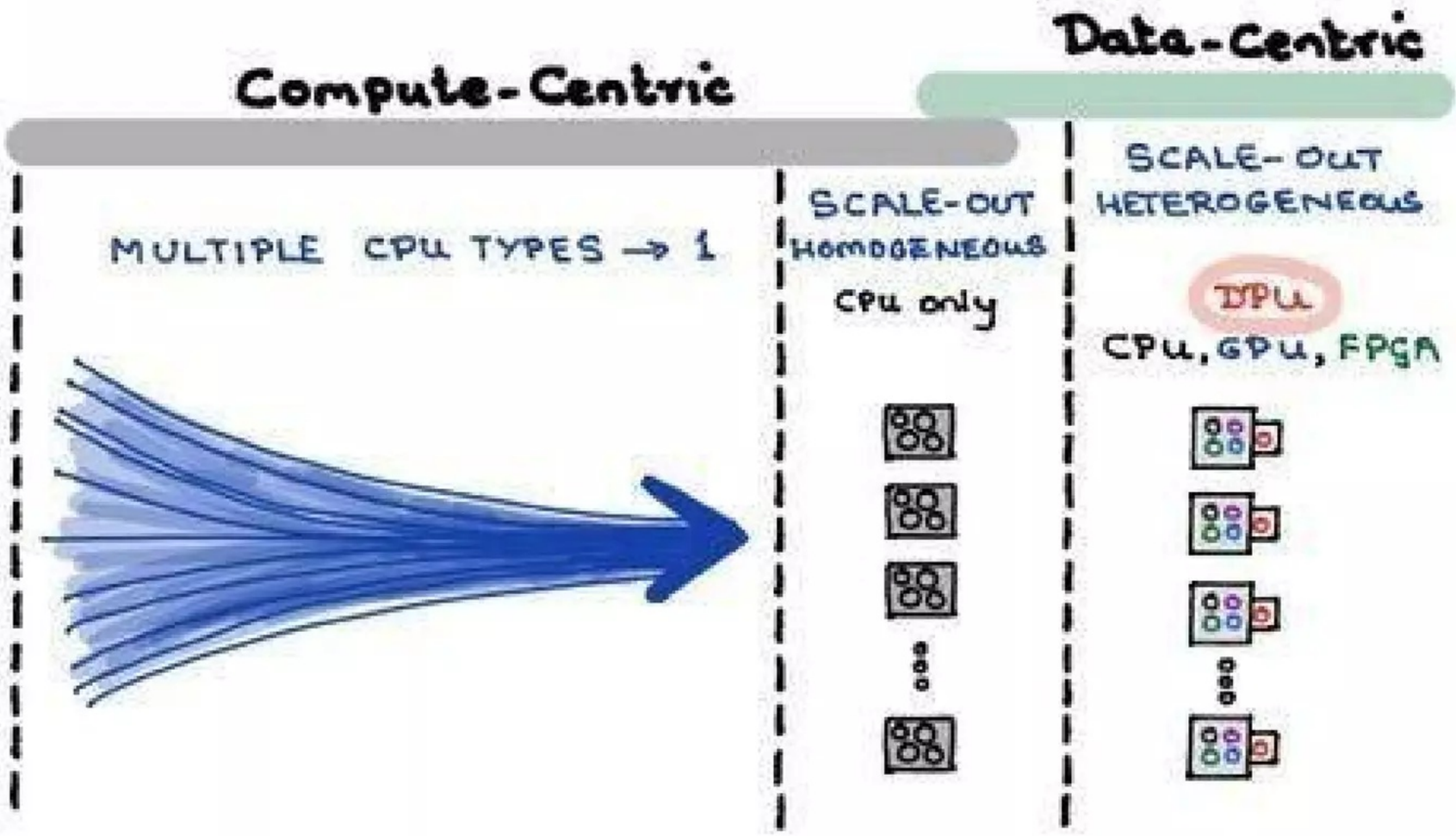




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FUNDAMENTAL INDUSTRY NEED FOR DPU

Modern applications: increasingly data-centric

Data centers: largely compute-centric

Data-Centric ≜
- workload arrives as packets
- client requests intermingled
- computation is stateful
- I/O dominates arithmetic

Data Processing Unit (DPU) is programmable silicon that:

- Executes data-centric workloads much more efficiently
- Enables disaggregation and pooling of compute and storage at scale

DPU represents new class of microprocessor focused on data-centric workloads

DPU will impact all levels of cloud infrastructure:

- Server
- D.C. Building
- Global

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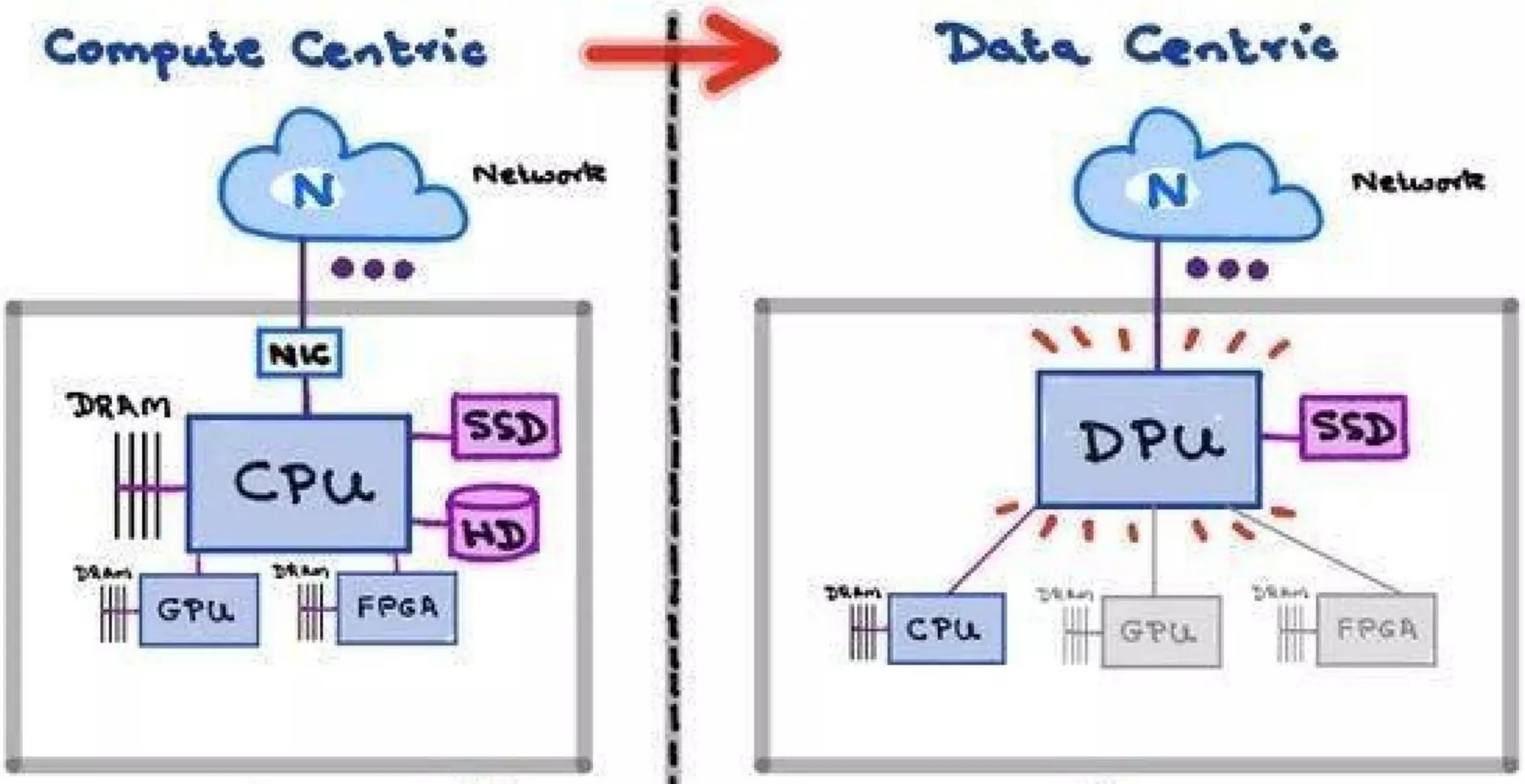
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Server Impact

90% of servers in 5 years



CPU: Apps ✓ + Traffic ✗ cop

GPU ✓, FPGA ✓: Applications

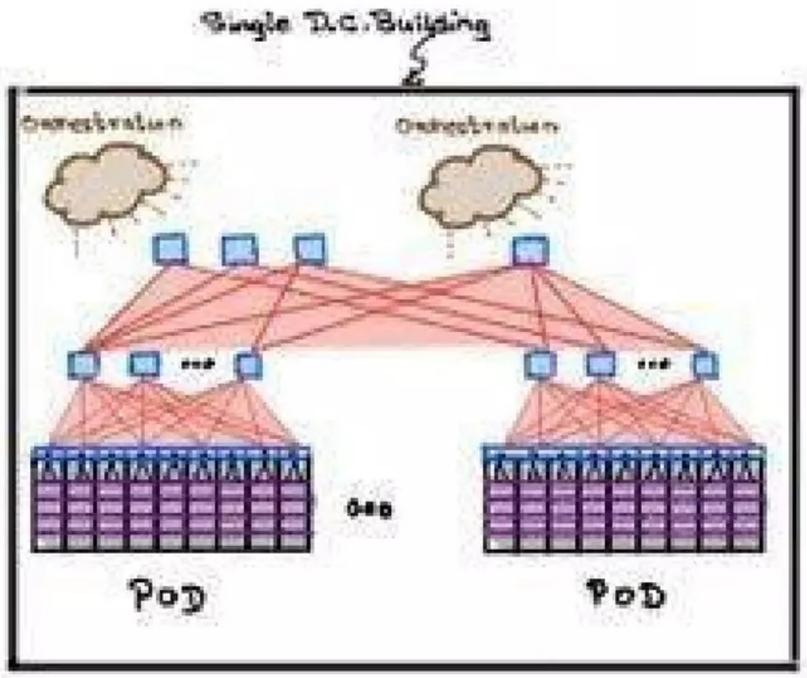
DPU: Data Centric ✓ Workload

CPU ✓, GPU ✓, FPGA ✓: Applications

DATA CENTER IMPACT

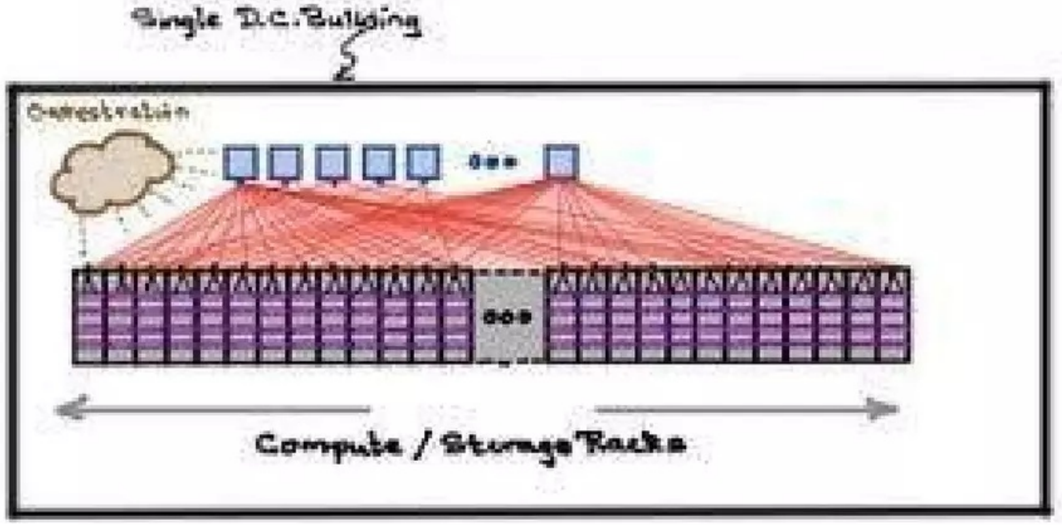
90% of buildings in 5 years

Complex multi-tier network



Multi-tier, inefficient, expensive slow

Simple flat fabric

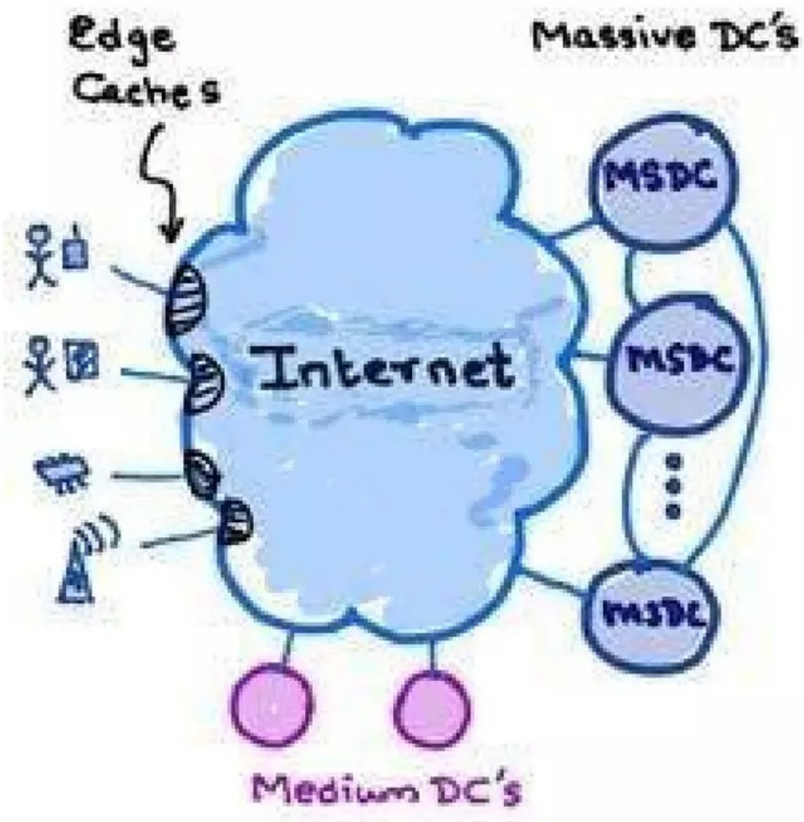


Flat, efficient, fast:
Supports Resource Pooling at large Scale

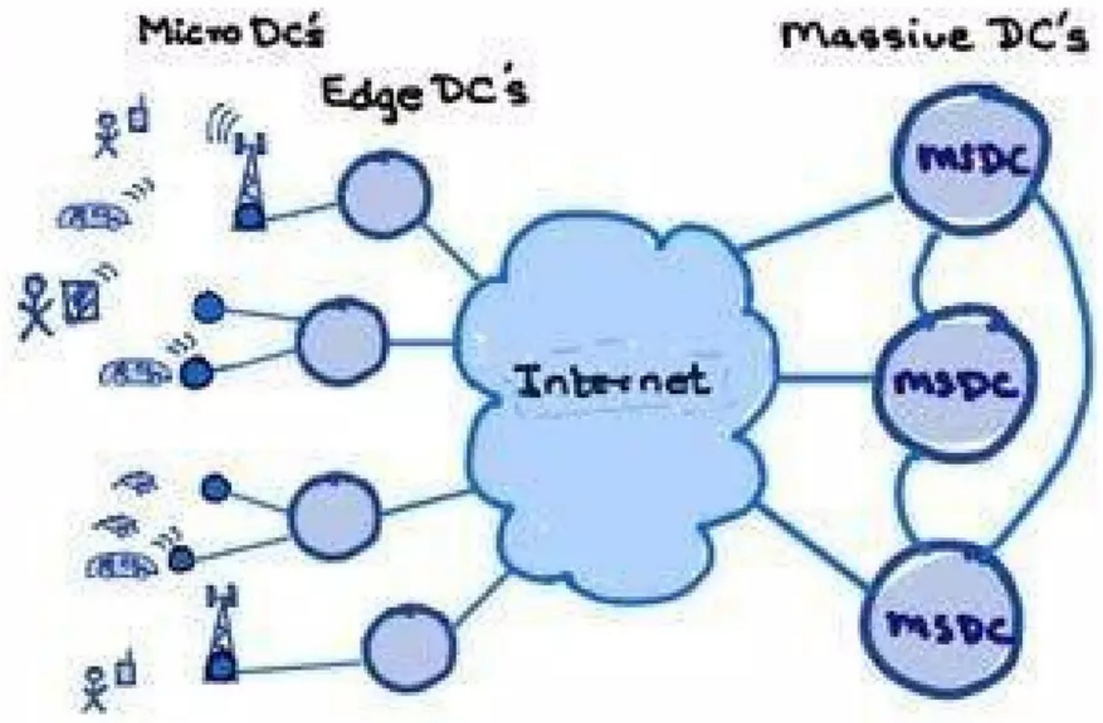
Global Impact

90% of regions in 5 years

Centralized



Hybrid



- Lower latency – AR, Siri, Gaming
- Lower cost
- Better security & durability
- Data Sovereignty

Summary:

- Fundamental new technology for a data-centric world
- Fully programmable DPU is a new class of microprocessor
  - + Performs data centric computations much more efficiently
  - + Enables disaggregation and pooling at scale
- Breakthrough benefits in:
  - Economics
  - Security
  - Reliability
  - Performance
  - Agility

  } Across all data centers

- Will enable Cloud Providers to leapfrog competition
- Designed for ease of insertion
  - No changes in Application code
  - No changes to network
  - No changes to server design

THANK You!

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